ChatGPT

What is the clinical significance of BCR-ABL1 Fusion AND ABL1 F359I in Chronic Myeloid Leukemia with Imatinib Mesylate in CIViC?

I'll look up recent and authoritative sources, such as CIViC (Clinical Interpretations of Variants in Cancer), to understand the clinical significance of the BCR-ABL1 Fusion and ABL1 F359I mutation in chronic myeloid leukemia when treated with Imatinib Mesylate. I'll compile the findings and share a detailed summary with you.

Worked for 6 minutes >

In Chronic Myeloid Leukemia (CML), the BCR-ABL1 fusion gene encodes a constitutively active ABL1 tyrosine-kinase. Imatinib mesylate targets this kinase; however, secondary mutations in the ABL1 kinase domain can disrupt drug binding and cause resistance. The F359I mutation (phenylalanine→isoleucine at codon 359) lies in the C-helix of ABL1 and is frequently detected in patients who fail imatinib therapy. Published clinical reports show that when this mutation arises in the BCR-ABL1 fusion, patients often fail to achieve a major molecular response on standard-dose imatinib; switching to dasatinib can restore response pubmed.ncbi.nlm.nih.gov. In one case report, a CML patient who did not respond to imatinib for 12 months was found to have an F359I mutation; nilotinib therapy produced rapid disease progression but changing to dasatinib led to clinical improvement pmc.ncbi.nlm.nih.gov. A review of resistance mutations lists F359I among single mutations that should exclude certain second-generation TKIs and notes that all mutations on the list (including F359I) are resistant to imatinib pmc.ncbi.nlm.nih.gov. A large compilation of kinase-domain mutations poorly sensitive to tyrosine-kinase inhibitors also lists F359I as poorly sensitive to both imatinib and nilotinib jhoonline.biomedcentral.com jhoonline.biomedcentral.com. Bosutinib sensitivity depends on the specific mutation; the F359I variant is categorized as bosutinib-insensitive with an IC50 > 2-fold higher than wild type pmc.ncbi.nlm.nih.gov.

**Clinical significance (CIViC interpretation)**

- **Evidence type – Predictive:** The F359I mutation predicts the patient's response to tyrosine-kinase inhibitors. Multiple studies and case reports have shown that CML patients with BCR-ABL1 fusion and F359I experience little benefit from imatinib mesylate and often develop rapid disease progression pmc.ncbi.nlm.nih.gov.
- **Clinical significance – Resistant:** F359I is a resistance mutation. It disrupts imatinib binding, leading to suboptimal response or treatment failure pubmed.ncbi.nlm.nih.gov pmc.ncbi.nlm.nih.gov. The mutation also confers resistance to nilotinib jhoonline.biomedcentral.com and reduced sensitivity to bosutinib pmc.ncbi.nlm.nih.gov.
- **Actionability:** Because of imatinib resistance, clinicians should avoid imatinib mesylate in patients with BCR-ABL1 F359I. Case reports and in vitro data suggest that dasatinib can overcome the F359I mutation; switching to dasatinib achieved complete cytogenetic and major molecular responses in patients who failed imatinib pubmed.ncbi.nlm.nih.gov. Bosutinib has limited activity against F359I, while ponatinib and allosteric inhibitors such as asciminib may retain activity, although data are limited.

In summary, CIViC classifies the BCR-ABL1 fusion with ABL1 F359I as a predictive biomarker of **resistance** to imatinib mesylate. The presence of this mutation should prompt clinicians to consider alternative tyrosine-kinase inhibitors (especially dasatinib) rather than continuing imatinib therapy.

Sources

Describe a task

Agent × Sources